

Ch38: 2, 9, 17, 25, 64

2. (a) Equation 38-11 yields

$$\Delta\lambda = \frac{h}{m_e c} (1 - \cos\phi) = (2.43 \text{ pm})(1 - \cos 180^\circ) = +4.86 \text{ pm}.$$

(b) Using the value $hc = 1240 \text{ eV} \cdot \text{nm}$, the change in photon energy is

$$\Delta E = \frac{hc}{\lambda'} - \frac{hc}{\lambda} = (1240 \text{ eV} \cdot \text{nm}) \left(\frac{1}{0.0085 \text{ nm} + 4.86 \text{ pm}} - \frac{1}{0.0085 \text{ nm}} \right) = -53.1 \text{ keV}.$$

(c) From conservation of energy, $\Delta K = -\Delta E = 53.1 \text{ keV}$.

(d) The electron will move straight ahead after the collision, since it has acquired some of the forward linear momentum from the photon. Thus, the angle between $+x$ and the direction of the electron's motion is zero.

9. (a) The transmission coefficient T for a particle of mass m and energy E that is incident on a barrier of height U_b and width L is given by

$$T = e^{-2bL},$$

where

$$b = \sqrt{\frac{8\pi^2 m (U_b - E)}{h^2}}.$$

For the proton, we have

$$\begin{aligned} b &= \sqrt{\frac{8\pi^2 (1.6726 \times 10^{-27} \text{ kg})(10 \text{ MeV} - 3.0 \text{ MeV})(1.6022 \times 10^{-13} \text{ J/MeV})}{(6.6261 \times 10^{-34} \text{ J} \cdot \text{s})^2}} \\ &= 5.8082 \times 10^{14} \text{ m}^{-1}. \end{aligned}$$

This gives $bL = (5.8082 \times 10^{14} \text{ m}^{-1})(10 \times 10^{-15} \text{ m}) = 5.8082$, and

$$T = e^{-2(5.8082)} = 9.02 \times 10^{-6}.$$

The value of b was computed to a greater number of significant digits than usual because an exponential is quite sensitive to the value of the exponent.

(b) Mechanical energy is conserved. Before the proton reaches the barrier, it has a kinetic energy of 3.0 MeV and a potential energy of zero. After passing through the barrier, the proton again has a potential energy of zero, thus a kinetic energy of 3.0 MeV.

(c) Energy is also conserved for the reflection process. After reflection, the proton has a potential energy of zero, and thus a kinetic energy of 3.0 MeV.

(d) The mass of a deuteron is $2.0141 \text{ u} = 3.3454 \times 10^{-27} \text{ kg}$, so

$$b = \sqrt{\frac{8\pi^2 (3.3454 \times 10^{-27} \text{ kg})(10 \text{ MeV} - 3.0 \text{ MeV})(1.6022 \times 10^{-13} \text{ J/MeV})}{(6.6261 \times 10^{-34} \text{ J}\cdot\text{s})^2}} \\ = 8.2143 \times 10^{14} \text{ m}^{-1}.$$

This gives $bL = (8.2143 \times 10^{14} \text{ m}^{-1})(10 \times 10^{-15} \text{ m}) = 8.2143$, and

$$T = e^{-2(8.2143)} = 7.33 \times 10^{-8}.$$

(e) As in the case of a proton, mechanical energy is conserved. Before the deuteron reaches the barrier, it has a kinetic energy of 3.0 MeV and a potential energy of zero. After passing through the barrier, the deuteron again has a potential energy of zero, thus a kinetic energy of 3.0 MeV.

(f) Energy is also conserved for the reflection process. After reflection, the deuteron has a potential energy of zero, and thus a kinetic energy of 3.0 MeV.

17. For $U = U_0$, Schrödinger's equation becomes

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2m}{h^2}[E - U_0]\psi = 0.$$

We substitute $\psi = \psi_0 e^{ikx}$. The second derivative is

$$\frac{d^2\psi}{dx^2} = -k^2\psi_0 e^{ikx} = -k^2\psi.$$

The result is

$$-k^2\psi + \frac{8\pi^2m}{h^2}[E - U_0]\psi = 0.$$

Solving for k , we obtain

$$k = \sqrt{\frac{8\pi^2m}{h^2}[E - U_0]} = \frac{2\pi}{h} \sqrt{2m[E - U_0]}.$$

25. If K is given in electron volts, then

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}} \Rightarrow K = \frac{h^2}{2m\lambda^2} = \frac{(6.626 \times 10^{-34} \text{ J}\cdot\text{s})^2}{2(9.109 \times 10^{-31} \text{ kg})(590 \text{ nm})^2} = 6.92 \times 10^{-25} \text{ J} = 4.33 \times 10^{-6} \text{ eV}$$

64. (a) For the first and second case (labeled 1 and 2) we have

$$eV_{01} = hc/\lambda_1 - \Phi, \quad eV_{02} = hc/\lambda_2 - \Phi,$$

from which h and Φ can be determined. Thus,

$$h = \frac{e(V_1 - V_2)}{c(\lambda_1^{-1} - \lambda_2^{-1})} = \frac{1.85 \text{ eV} - 0.820 \text{ eV}}{(3.00 \times 10^{17} \text{ nm/s})[(300 \text{ nm})^{-1} - (400 \text{ nm})^{-1}]} = 4.12 \times 10^{-15} \text{ eV}\cdot\text{s}.$$

(b) The work function is

$$\Phi = \frac{hc}{\lambda_1} - V_1 = \frac{hc}{\lambda_2} - V_2 = 2.27 \text{ eV}.$$

(c) Let $\Phi = hc/\lambda_{\text{max}}$ to obtain

$$\lambda_{\text{max}} = \frac{hc}{\Phi} = \frac{1240 \text{ eV}\cdot\text{nm}}{2.27 \text{ eV}} = 545 \text{ nm}.$$